The Social Inevitability of Personal Irreducibility in the Age of AI: A Theoretical Analysis of Responsibility-History Structures and the Critical Point of Semantic Decay

Yuto Ogawa
Independent Researcher
ORCID: 0009-0003-8419-2854
Email: yuto0804sun@gmail.com

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Abstract

This paper analyzes the most essential social impact of rapid AI development not as labor substitution or mere information-generation capacity, but as the collapse and reconfiguration of responsibility structures. In traditional societies, social decision-making has been sustained by the coupling of meaning and responsibility attribution (i.e., who irreversibly bears the consequence). However, AI separates information from subjects through unbounded generativity, imitability, and context-stripping, thereby decoupling meaning from responsibility. As a consequence, society moves toward a *critical point of semantic decay* while simultaneously re-demanding *personal irreducibility*—a fixed point defined by irreversible responsibility histories.

We define personal irreducibility as the social inevitability of irreversible responsibility histories and present: (i) a formalization of semantic decay, (ii) a staged model of early signals leading to the critical point, (iii) necessary and sufficient conditions under which personal irreducibility emerges as a fixed point, and (iv) the role of recording infrastructures (date-numbering, cryptographic hashes, DOI-like identifiers) as a coordinate system that makes responsibility histories publicly addressable. We conclude that the core problem of the AI era is not information generation but how to reconstruct irreversible structures of responsibility attribution.

Structure of the Paper

Chapter 1 formalizes the problem setting: AI disrupts social decision-making by separating meaning and responsibility. Chapter 2 defines personal irreducibility as a responsibility-history fixed point and distinguishes it from ability, trust, and brand. Chapter 3 derives semantic decay from core AI properties and describes the dynamics toward the critical point. Chapter 4 organizes early signals as a staged model. Chapter 5 specifies the conditions under which personal irreducibility becomes socially inevitable. Chapter 6 presents recording infrastructures as a coordinate system that preserves irreversible responsibility histories. Chapter 7 concludes that the re-demand for personal irreducibility is inevitable in the age of AI.

1 Introduction: Responsibility Structures as the Core Problem of the AI Era

Discussions of AI development have largely concentrated on labor substitution, economic impacts, ethical risks, or truth verification. Yet these concerns are surface-level effects. The fundamental issue is that AI transforms the coupling condition between *meaning* and *responsibility* in society, thereby disrupting the very basis on which decisions become socially realizable.

In traditional societies, discourse, judgment, and knowledge have been implicitly bound to a responsible subject. Social decision-making is not established merely by the existence of information; it requires the simultaneous establishment of meaning and responsibility attribution:

$$\text{Dec} \iff (\text{Meaning}) \wedge (\text{Attr}). \tag{1}$$

Here, responsibility attribution refers to the existence of an *attribution point* that can bear the burden of failure, loss, or irreversible consequence.

AI, however, detaches information from subjects and converts it into a state of unbounded generation and replication. Society thus confronts a structural change in which meaning and responsibility become decoupled:

$$(\text{Meaning}) \perp (\text{Resp}). \tag{2}$$

This is not merely an increase in misinformation. Rather, it dissolves the attribution points that societies have relied upon to finalize decisions.

This paper formalizes this transformation as a **semantic decay structure** and introduces the **critical point of semantic decay** as its limiting regime. Crucially, this critical point is not only a collapse but also a process through which society re-demands a responsibility fixed point—namely,

$$\text{personal irreducibility (a responsibility-history fixed point).} \tag{3}$$

We show that personal irreducibility becomes **socially inevitable** precisely because decision-making requires an irreversible attribution point.

Contributions and scope (clarified without shortening)

To preserve clarity while increasing detail, we state the contributions in a more explicit operational form. This paper contributes:

1. A *structural* formalization of the AI-driven separation of meaning and responsibility (not a moral critique).

2. A *definition* of personal irreducibility as the social inevitability of irreversible responsibility histories, and a strict separation of this concept from ability, trust-as-accuracy, and brand.

3. A *staged model* of early signals that appear *before* collapse, including the precise moment at which the re-demand for irreducibility rises.

4. A *coordinate-system* account of recording infrastructures (date-numbering, hashes, DOI-like identifiers) as the mechanism that preserves the public referability of responsibility histories across time.

5. Two supplementary propositions that (i) formalize the *final reduction structure* of attribution to an irreducible person and (ii) derive the *inevitability* of premise-updating capability as the only structural exit from semantic decay.

The next chapter removes ambiguity from the notion of personal irreducibility by defining it rigorously as a responsibility-history structure.

2 A Theoretical Definition of Personal Irreducibility: The Subject as a Responsibility-History Fixed Point

This chapter defines personal irreducibility (authenticity / personal irreducibility) in the AI era as a strict structural requirement for social decision-making, independent of popular notions such as ability, trust, or brand. In this paper, personal irreducibility is the state in which irreversible responsibility histories accumulate such that social decisions cannot be finalized without passing through that subject.

2.1 Limits of conventional notions: confusion with ability, trust, and brand

Personal irreducibility has often been conflated with individuality, talent, expertise, trust relations, or brand value. Yet all such properties remain socially substitutable and are particularly unstable in the AI era.

Ability. Ability can be surpassed and replicated by AI. If personal irreducibility is defined as superior ability,

$$\text{personal irreducibility} \equiv \text{ability}, \tag{4}$$

then it loses meaning as AI performance increases. This is not a prediction about *all* skills being replaced at once, but a structural statement: ability is a *comparative variable* and therefore cannot serve as a stable fixed point under a regime of rapid capability scaling.

Trust (misreduced to accuracy). Trust can migrate to AI when it is reduced to outcome accuracy. If trust is mistaken as "being correct,"

$$\text{trust} \rightarrow \text{accuracy maximization}, \tag{5}$$

then delegating to higher-accuracy predictors becomes rational. The crucial observation is that "trust" in this reduction ceases to be a social attribution structure and becomes a performance metric; once that happens, the trust relation is technically transferable.

Brand. Brand is a social consensus artifact and is fragile under imitation and synthetic generation. The core issue is not simply that these collapse, but that the *minimum structure required to continue social decision-making* becomes exposed. When imitation becomes cheap, brand loses its ability to guarantee an attribution point unless it is backed by a non-bypassable responsibility history.

2.2 Definition: the social inevitability of irreversible responsibility histories

We define personal irreducibility as follows:

> **Personal irreducibility is the state in which irreversible decision histories accumulate such that social decision-making cannot be finalized without passing through that subject.**

This definition is not about "uniqueness" or "originality" but about an **attribution fixed point** required for decisions. A social decision requires:

$$\text{Dec} \Rightarrow \exists\, \text{responsible subject } R \text{ s.t. } R \text{ becomes an attribution point.} \tag{6}$$

This point is not merely "who said it," but the locus where irreversible commitments accumulate and become socially non-bypassable.

2.3 Conditions for establishment: irreversibility, social impact, historicity

Personal irreducibility requires three conditions:

1. **Irreversibility**: decisions are socially non-retractable.
2. **Social impact**: decisions constrain or re-shape others' choice spaces.
3. **Historicity**: the above accumulate over time as a referable structure.

Let the subject's decision sequence be

$$D_R = \{d_1, d_2, \ldots, d_n\}, \tag{7}$$

with irreversibility modeled as

$$d_k \not\equiv \varnothing, \qquad d_k \text{ is socially non-retractable.} \tag{8}$$

Let others' action set be A_{others}. Social impact means the decision history constrains that space:

$$A_{\text{others}} \xrightarrow{D_R} A'_{\text{others}}, \qquad |A'_{\text{others}}| < |A_{\text{others}}|. \tag{9}$$

Historicity means personal irreducibility is not a momentary attribute but a history function:

$$\text{personal irreducibility} = \Phi(D_R(t)). \tag{10}$$

2.4 Consistency with FCT: personal irreducibility as a social future fixed point

Within Future Convergence Theory (FCT), decision-making is described as convergence toward a future fixed point $\Lambda(T)$. In an AI-saturated environment, the decisive factor becomes which attribution point society must pass through to finalize decisions. Thus personal irreducibility functions as a social fixed point:

$$\Lambda_{\text{social}} \equiv R_{\text{fixed}}. \tag{11}$$

The key asymmetry is that AI can generate content but cannot accumulate irreversible "bearing" as a socially recognized attribution point.

2.5 Conclusion of this chapter

We defined personal irreducibility as the social inevitability of irreversible responsibility histories, distinct from ability, trust, or brand. The next chapter derives semantic decay from AI's core structural properties.

3 AI and the Semantic Decay Structure: A Generative Civilization as Responsibility Separation

This chapter derives how AI's structural properties separate meaning from responsibility, inevitably producing a *semantic decay structure*. We do not morally condemn AI; rather, we treat it as a device that exposes the responsibility structure implicit in society.

3.1 Core properties: unbounded generativity, perfect imitability, context-stripping

Generative AI exhibits three essential properties:

1. **Unbounded generativity**
2. **Perfect imitability**
3. **Context-stripping capability**

Unbounded generativity. Unbounded generativity yields a regime in which generation flow G approaches infinity:

$$G \to \infty. \tag{12}$$

This is not a literal claim about physical infinity, but a structural claim: marginal production cost approaches zero while output scale approaches the reference capacity of society.

Perfect imitability. Perfect imitability yields *source isomorphism*. If a subject's discourse style is S_R, then:

$$\exists\,\mathrm{AI} \ \text{ s.t. } \ S_{\mathrm{AI}} \approx S_R. \tag{13}$$

This creates the social condition where "who said it" becomes informationally unreliable unless supported by separate identity-binding mechanisms.

Context stripping. Context stripping detaches the responsibility-bearing history behind discourse and allows meaning to circulate as independent symbols:

$$\text{Meaning} = (\text{Symbol}) \ominus (\text{Responsibility context}). \tag{14}$$

Here, $\ominus$ denotes that responsibility context can be subtracted from circulation while leaving surface semantic form intact.

3.2 Definition of semantic decay: separation of discourse from responsibility attribution

We define semantic decay as:

> **Semantic decay is the state in which discourse separates from responsibility attribution, and discourse no longer requires a subject to function socially.**

This is deeper than misinformation. In traditional settings, discourse X contributes to decision-making only if responsibility can be mapped to a subject:

$$X \Rightarrow \exists\, R \ \text{ s.t. } \ \mathrm{Resp}(X) \mapsto R. \tag{15}$$

Under AI-dominated generation, discourse increasingly circulates as attribution-null:

$$X \Rightarrow R = \varnothing, \tag{16}$$

where $R = \varnothing$ denotes not anonymity but structural non-attribution. When this proliferates:

$$\forall X, \ \ \mathrm{Resp}(X) \not\mapsto \mathrm{Fix}, \tag{17}$$

society loses responsibility fixed points.

3.3 Dynamics: generation explosion relativizes attribution

Let the finite discourse set be $\mathcal{X}$ and source metadata be $\sigma(X)$. Traditional evaluation is:

$$\mathrm{Dec} = \mathcal{D}(\mathcal{X}, \sigma). \tag{18}$$

As AI expands discourse supply,

$$|\mathcal{X}| \to \infty, \tag{19}$$

the finite capacity of reference weakens σ, yielding pressure toward:

$$\mathcal{D}(\mathcal{X}, \sigma) \to \mathcal{D}(\mathcal{X}). \tag{20}$$

This appears rational but structurally erases the attribution points required for decisions.

3.4 What this paper does *not* claim

We do not claim:

1. "AI destroys meaning, therefore it should be banned."
2. "AI is ethically evil."
3. "All generated information is worthless."

We claim only that AI structurally separates meaning from responsibility and that beyond a threshold, this drives decision-making into a critical regime.

3.5 Consistency with FCT: semantic decay as social fixed-point loss

In FCT, decision-making is convergence toward fixed points. Semantic decay corresponds to the loss of the social responsibility fixed point:

$$\Lambda_{\text{social}}\ (\text{responsibility fixed point}) \to \varnothing. \tag{21}$$

Thus an increase in candidate answers can paradoxically produce non-convergence:

$$\text{more answers} \Rightarrow \text{less convergence}. \tag{22}$$

3.6 Conclusion of this chapter

AI's unbounded generation, imitability, and context stripping detach discourse from subjects, producing semantic decay as the loss of responsibility fixed points. The next chapter organizes the early signals observed as society approaches the critical point.

4 Early Signals of the Critical Point: The First Observable Rise of the Re-demand for Irreducibility

This chapter organizes early signals that arise as semantic decay progresses and society approaches the critical point at which personal irreducibility becomes an inevitable re-demand. "Signals" here are not post-collapse outcomes but *structural changes observable during approach.*

4.1 Definition of the critical point of semantic decay

Semantic decay progresses continuously, but there exists a critical point at which social decision-making phase-transitions into dysfunction. We call this the *critical point of semantic decay* (CPSD).

To prevent overflow, we express the defining conditions in a forced multi-line form:

$$\begin{aligned} \text{(i) Meaning flow:} \quad & G \to \infty, \\ \text{(ii) Attribution fixed point:} \quad & R \to \varnothing, \\ \text{(iii) Decision realizability:} \quad & \mathrm{P}(\mathcal{D}) \to 0. \end{aligned} \tag{23}$$

Here $\mathrm{P}(\mathcal{D})$ is an abstract measure of the probability that society can finalize a decision. The critical point is not merely "more information," but a phase transition into non-finalization.

4.2 A four-stage model of progression

We organize progression into four stages:

1. Source neutralization
2. Authority dilution
3. Decision paralysis
4. Responsibility vacuum (critical regime)

These are not moral decline but structural consequences of fixed-point loss.

4.3 Stage 1: Source neutralization

Traditionally, social value depended on both content X and source $\sigma(X)$:

$$V(X) = \Phi(X, \sigma(X)). \tag{24}$$

Under source isomorphism,

$$\sigma(X) \approx \sigma(Y) \quad \text{for many } X \neq Y, \tag{25}$$

pressure arises toward:

$$V(X) = \Phi(X, \sigma(X)) \to V(X) = \Phi(X). \tag{26}$$

The first critical signal is the spontaneous emergence of the statement:

> "Ignore who said it; evaluate only the content."

Structurally, this legitimizes the deletion of responsibility fixed points.

4.4 Stage 2: Authority dilution

Experts historically carried authority partly because they bore responsibility. Under AI:

$$A_{\text{AI}} \geq A_E, \qquad R_{\text{AI}} = \varnothing, \tag{27}$$

so the binding force of expert judgment declines:

$$\text{binding force}(E) \to 0. \tag{28}$$

This is not primarily about reduced competence, but about responsibility becoming visible as a cost.

4.5 Stage 3: Decision paralysis

Let AI supply an option set $\mathcal{A}$:

$$|\mathcal{A}| \to \infty. \tag{29}$$

Yet decision-making requires a cut—an irreversible "we choose this." Cuts require a responsibility fixed point. Without it, the decision operator becomes structurally undefined:

$$\mathcal{D}(\mathcal{A}) \text{ is not well-defined.} \tag{30}$$

4.6 Stage 4: Responsibility vacuum

At the critical regime, responsibility cannot be mapped:

$$\forall X, \;\; \text{Resp}(X) \not\mapsto R^*. \tag{31}$$

Society then tends toward either:

$$\begin{aligned} &\text{(A) Formal decisions (decisions without responsibility),} \\ \text{or} \quad &\text{(B) Reintroduction of fixed points (re-demand for irreducibility).} \end{aligned} \tag{32}$$

We argue that (A) lacks long-run stability, so society is structurally attracted to (B).

4.7 When the re-demand rises (moment of inevitability)

The re-demand for personal irreducibility rises precisely when:

$$\Big(\text{a cut is required to decide}\Big) \ \wedge \ \Big(\text{no responsibility fixed point exists}\Big). \tag{33}$$

The experience of "we must decide but cannot finalize" itself summons irreducibility as an inevitable structure.

4.8 Conclusion of this chapter

We presented a staged model of early signals and showed that the first observable signal is source neutralization. The next chapter formalizes why personal irreducibility must re-emerge as a responsibility fixed point after the critical point.

5 The Social Inevitability of Personal Irreducibility: Re-convergence as a Responsibility Fixed Point

This chapter formalizes why, after the critical point, society must reintroduce personal irreducibility as a *responsibility fixed point*. AI-driven semantic decay does not eliminate irreducibility; it exposes that irreducibility is a condition for decision formation.

5.1 Decision formation requires a fixed point

Let options be $\mathcal{A}$, social state be $S(t)$, and decision operator be $\mathcal{D}$:

$$\mathcal{D}: \ (\mathcal{A}, S(t)) \mapsto a^* \in \mathcal{A}, \qquad S(t+1) = \Psi(S(t), a^*). \tag{34}$$

With AI,

$$|\mathcal{A}| \to \infty, \tag{35}$$

so decision formation depends less on computation and more on the *ground of binding*.

5.2 Responsibility fixed point

Define the responsibility fixed point R^* as:

> R^* is the reference point to which the consequence of a decision a^* irreversibly maps, and without which society cannot finalize decision-making.

Hence, social decision formation must output not only a^* but also R^*:

$$(\mathcal{A}, S(t)) \xrightarrow{\mathcal{D}} (a^*, R^*). \tag{36}$$

If R^* is absent, a^* loses binding force as "nobody's decision."

5.3 The paradox: as content becomes non-scarce, attribution becomes decisive

AI can supply unlimited content-equivalents. Thus value shifts from content superiority to attribution structure:

$$V(X) \text{ depends on content } \rightarrow V(X) \text{ depends on attribution.} \tag{37}$$

The structural paradox is:

> The more AI eliminates content scarcity, the more the scarcity of responsibility fixed points becomes socially amplified.

5.4 Irreducibility as irreversible accumulation

Let subject i have responsibility history $H_i(t)$, and let $\Omega(a^*)$ denote the irreversible consequence mapping:

$$H_i(t+1) = H_i(t) \cup \Omega(a^*). \tag{38}$$

Irreducibility is established when:

$$\exists i \text{ s.t. } \forall \mathcal{D},\ \mathcal{D} \text{ cannot be finalized without referencing } i. \tag{39}$$

Then i is a social fixed point.

5.5 Precise inevitability conditions

Inevitability rises when the following co-occur:

> Truth assessment becomes unstable: $\mathrm{P}(\text{true} \mid X)$ does not stabilize,
> Authority binding collapses: binding force$(E) \rightarrow 0$,
> Attribution mapping disappears: $X \mapsto R^*$ is not well-defined.

Then decision realizability collapses:

$$\mathrm{P}(\mathcal{D}) \rightarrow 0, \tag{40}$$

forcing society toward either responsibility-free formalism or reintroduction of fixed points.

5.6 Consistency with FCT: fixed points as the social counterpart of future convergence

To avoid overflow, we define x^* and R^* in a two-line aligned form:

$$\begin{aligned} x^* &: \text{convergence point of an individual's choice space,} \\ R^* &: \text{convergence point of society's responsibility space.} \end{aligned} \tag{41}$$

Thus irreducibility is the responsibility-coordinate by which society reintroduces a future fixed point.

5.7 Conclusion of this chapter

After the critical point, society must reintroduce responsibility fixed points. Personal irreducibility is not "ability against AI," but the irreversible history needed to make decisions bind. The next chapter specifies the recording infrastructure required to preserve such histories over time.

6 Recording Infrastructure: A Reference Coordinate System that Transports Responsibility Histories into the Future

This chapter distinguishes personal irreducibility itself from the recording infrastructure that preserves its referability across time. The infrastructure does not *create* irreducibility; it ensures that an already-established responsibility history remains publicly addressable.

6.1 Irreducibility is not identical to records

$$\text{personal irreducibility} \neq \text{recording infrastructure}, \qquad \text{but} \quad \text{infrastructure} \Rightarrow \text{public referability}. \tag{42}$$

6.2 Three requirements (expanded as design constraints)

A responsibility history must satisfy:

1. **Temporal ordering**: the history must preserve a verifiable sequence of commitments.
2. **Identity binding**: the history must be bound to a subject in a way that makes substitution detectable.
3. **Public addressability**: the history must be referable by third parties in shared decision spaces.

These are not merely "documentation best practices." In this paper they are treated as *structural necessities* for the persistence of attribution fixed points.

6.3 Temporal ordering

Let history elements be h_k with order $\prec$:

$$h_1 \prec h_2 \prec \cdots \prec h_n. \tag{43}$$

Date-numbering and versioning are external representations of this ordering, but become robust only when combined with identity binding. The reason is structural: ordering alone can be forged; ordering plus binding makes forgery detectable.

6.4 Identity binding

Because content is replicable, "same content" no longer proves attribution. Identity binding cryptographically binds h_k to subject i:

$$\mathcal{B} : (i, h_k) \mapsto \sigma_{i,k}, \tag{44}$$

where $\sigma_{i,k}$ includes signatures, hashes, and public-key proofs. If h_k is altered, the binding becomes inconsistent:

$$h_k \text{ modified } \Rightarrow \sigma_{i,k} \text{ inconsistent.} \tag{45}$$

In other words, identity binding is not a claim about identity but an externally verifiable failure mode when tampering occurs.

6.5 Public addressability

Responsibility histories must be third-party referable. Let the public address be $A(h_k)$:

$$\mathcal{P} : h_k \mapsto A(h_k), \tag{46}$$

where $A(h_k)$ can be DOI-like persistent identifiers, archive IDs, or persistent URLs. Public addressability is what moves a responsibility history from private memory into the shared reference space in which decisions are negotiated.

6.6 Reference chain

Combining the three yields a reference chain:

$$h_k \xrightarrow{\text{order}} t_k \xrightarrow{\text{bind}} \sigma_{i,k} \xrightarrow{\text{public}} A(h_k). \tag{47}$$

As long as this chain holds, "who, when, what, and under what responsibility" does not break even under unbounded AI generation.

6.7 Limit: infrastructure does not substitute irreducibility

$$\text{infrastructure} \not\Rightarrow \text{automatic generation of irreducibility.} \tag{48}$$

The infrastructure preserves and transports; it does not itself create the irreversible bearing that constitutes irreducibility.

6.8 Consistency with FCT: externalization of a future fixed point

In FCT, x^* governs convergence. Recording infrastructure externalizes the convergence so the social space can refer to R^* through $A(h_k)$.

6.9 Conclusion of this chapter

We formalized the infrastructure as a coordinate system that transports responsibility histories into the future via ordering, identity binding, and public addressability. The final chapter states the concluding thesis: AI exposes irreducibility rather than erasing it.

7 Conclusion: AI Does Not Erase Personal Irreducibility—It Exposes It

7.1 Main results

We redefined the core problem of the AI era as the collapse and reconfiguration of the coupling between meaning and responsibility. The main results are:

1. Personal irreducibility is not ability, trust, or brand, but the state in which irreversible responsibility histories become the fixed point required for social decision-making.
2. AI accelerates the separation of information from responsibility through unbounded generation, imitability, and context stripping, thereby inducing semantic decay.
3. Semantic decay progresses in stages and reaches a critical point at which responsibility attribution becomes structurally hollow.
4. At the critical point, decision-making cannot be sustained without reintroducing responsibility fixed points.
5. To transport responsibility histories into the future, a reference chain is required: temporal ordering, identity binding, and public addressability.

7.2 Final thesis (formalized without overflow)

AI does not eliminate personal irreducibility; via the critical point of semantic decay, it exposes irreducibility as the structural condition for social decision formation.

We express the thesis as a multi-line implication chain:

$$\begin{aligned}\text{Unbounded generation (AI)} &\Longrightarrow \text{Separation of meaning and responsibility}\\ &\Longrightarrow \text{Progression of semantic decay}\\ &\Longrightarrow \text{Critical point of semantic decay}\\ &\Longrightarrow \text{Re-demand for responsibility fixed points}\\ &\Longrightarrow \text{Social inevitability of personal irreducibility.}\end{aligned}$$

7.3 Consistency with FCT

In FCT, understanding and decision-making are established by sharing future fixed points, not by matching information volumes. AI supplies massive past and generated information but does not automatically supply a future fixed point. Instead, option overflow and responsibility hollowing drive society toward the critical point. There, irreducibility functions as the responsibility fixed point that reintroduces a future fixed point into the social system.

7.4 Practical implication

What remains as value in the AI era converges to one point:

> **An entity that bears irreversible responsibility histories and continuously presents a future fixed point.**

Date-numbering, cryptographic hashes, and DOI-like identifiers do not create irreducibility; they act as a coordinate system that keeps irreducibility referable in the future.

7.5 Future work

Future work includes:

1. Empirical indicators of the critical point (social phenomena, behavioral data, institutional changes)
2. Design of robust recording infrastructures (tamper resistance, durability, public addressability)
3. Concrete institutional models connecting irreducibility to decision processes (contracts, governance, education)

Supplementary Proposition 1: Final Reduction Structure of Responsibility Attribution

Even when institutions, collectives, or AI mediate responsibility, social attribution must reduce to an irreducible personal subject. To increase clarity without shortening, we restate the proposition as a structural lemma and its consequence.

Lemma 1 (Decision formation requires attribution)

Decision formation requires:

$$\mathrm{Dec} \iff (\mathrm{choice}) \wedge (\mathrm{Attr}). \tag{49}$$

That is, a socially binding decision is not merely a selection of an option; it is a selection together with a terminal attribution point that can bear irreversible consequence.

Lemma 2 (Existence of a responsibility fixed point)

Hence for any decision there must exist a responsibility fixed point:

$$\exists R^* \text{ s.t. } \mathrm{Resp}(a^*) \mapsto R^*. \tag{50}$$

Mediation vs. terminal bearing

Institutions and AI can mediate attribution (e.g., distribute procedures, audits, approvals, or formal accountability), but they cannot internally bear irreversible consequence as the terminal point in the same sense as a person does within a social recognition regime. Therefore attribution reduces along a chain:

$$a^* \mapsto R_{\text{institution}} \mapsto R_{\text{collective}} \mapsto R_{\text{individual}}. \tag{51}$$

Proposition 1 (Final reduction)

Thus:

Social responsibility attribution ultimately reduces to an irreducible personal subject. (52)

As semantic decay progresses, this final reduction structure becomes increasingly visible, because mediation layers become increasingly easy to imitate, automate, or disclaim.

We summarize the refined core chain as:

$$\begin{aligned}\text{AI-driven semantic decay} &\Longrightarrow \text{Loss of responsibility fixed points}\\ &\Longrightarrow \text{Exposure of the final reduction structure}\\ &\Longrightarrow \text{Social inevitability of personal irreducibility.}\end{aligned}$$

Supplementary Proposition 2: The Inevitability of Premise-Updating Capability

From the foregoing:

$$\text{Dec} \Longleftrightarrow (\text{choice}) \wedge (\text{Attr}). \tag{53}$$

In the AI era:

$$\begin{aligned}\text{loss of responsibility fixed points} &\Longrightarrow \text{hollowing-out of meaning}\\ &\Longrightarrow \text{structural non-formation of decision-making.}\end{aligned}$$

Crucially, this arises because current decision systems are designed as *optimization under fixed premises*:

$$\text{correctness} = \text{optimization within fixed premises.} \tag{54}$$

However, reconstructing responsibility attribution is not achievable inside fixed premises, because responsibility attribution depends on which future is *assumed and borne*—i.e., the selection of premises themselves.

Key claim (reconstructed with intentional line breaks; Eq. (55) repaired)

Hence:

A decision-making system that lacks the capability to update premises cannot reconstruct meaning and responsibility attribution in a semantic-decay regime, because the reconstruction requires selecting (and bearing) the premises that define which future the decision is "correct" for. (55)

Therefore:

$$\text{avoidance of semantic decay} \Longleftrightarrow \text{existence of premise-updating capability.} \tag{55}$$

We summarize:

$$\begin{aligned}\text{AI-driven semantic decay} &\Longrightarrow \text{loss of responsibility fixed points}\\ &\Longrightarrow \text{non-formation of the current premise system}\\ &\Longrightarrow \text{inevitability of premise-updating capability.}\end{aligned}$$